

Krrishpana Karmacharya

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Summary

AI/ML undergrad with experience building and deploying LLM applications, from RAG-based assistants to MLOps pipelines. Comfortable working across full stack, including data pipelines, model integration and API design. Currently diving deeper into agentic AI architectures and how to make these AI systems more robust.

Skills

- **Languages:** Python, SQL
- **AI & Machine Learning:** Prompt Engineering, Retrieval-Augmented Generation (RAG), Scikit-learn, MLflow, Pandas, NumPy
- **Backend & Web:** Django, FastAPI
- **Data & Infrastructure:** Docker, MariaDB, Redis, Apache Airflow, YAML, Git
- **Soft Skills:** Problem-solving, Team Collaboration, Adaptability, Learning Agility

Projects

Agentic AI Academic Assistant

AI-Academic-Agent

Tech Stack: Python, Google Gemini API, LangChain, ChromaDB, Pydantic, JSON, Git

- Built a Gemini-powered academic assistant with persistent conversation history, configurable personas, and streaming responses, handling malformed API responses with solid error handling.
- Implemented RAG-based document retrieval with ChromaDB and built a lightweight evaluation pipeline to measure retrieval relevance, guiding iterative improvements to chunking and retrieval quality.
- Added a guardrails layer with Pydantic-enforced output schemas to filter unsafe inputs and validate model outputs before use.
- Designed a modular, environment-configured architecture with JSON-based memory management.

Bytebites - Nutrition Based Planner – 3rd Place, Innovation Fest

Nutrition-Planner

Tech Stack: Python, Django, Google Gemini API (text + vision), HTML/CSS/JS

- Built a Django-based nutrition platform tailored to Nepali dietary habits, using a curated local food dataset to generate personalized meal plans based on user health profiles (BMI, BMR, medical history).
- Implemented image-based calorie estimation using Gemini's multimodal vision capabilities, allowing users to upload meal photos for automatic ingredient identification and macronutrient analysis.
- Developed progress-tracking features with JSON API endpoints for real-time weight and calorie visualizations, alongside an appointment booking system for health advisors.

Online Retail MLOps

Retail-MLOps

Tech Stack: Apache Airflow, MariaDB, Great Expectations, Pandas, Redis, Scikit-learn, XGBoost, MLflow, FastAPI, Docker, Evidently AI

- Designed and orchestrated an end-to-end MLOps pipeline with Apache Airflow for retail demand forecasting, automating ETL for the Online Retail II dataset into a Star Schema OLAP warehouse (MariaDB ColumnStore) for feature engineering.
- Enforced data quality with Great Expectations validation checks and built high-performance preprocessing pipelines using Pandas, PyArrow, and Redis caching ahead of model training.
- Trained and tracked demand-forecasting models (Scikit-learn, XGBoost) with MLflow for experiment tracking and versioning, then deployed via FastAPI and Docker for real-time serving.
- Implemented continuous monitoring with Evidently AI to detect data and model drift and trigger retraining, completing a full MLOps lifecycle.

Leadership & Hackathons

Joint Secretary, Sunway Student Representative Council Feb 2025 - Jan 2026

- Managed council operations and official documentation, streamlining administrative workflows for the student body.
- Coordinated and executed student events, strengthening cross-team communication and organizational logistics.

Organizing Committee Member, Codarambha 2025

- Facilitated cross-team coordination and documentation for a college-wide hackathon.
- Tracked project progress and documented key event milestones throughout the competition.

Organizing Committee Member & Emcee, AIDA 2025 2025

- Co-organized and emceed an AI-focused accessibility event, ensuring smooth program execution.
- Worked closely with mentors, judges, and the organizing team to ensure the event ran smoothly and successfully.

Participant, Hack for Nepal 2082 2025

- Rapidly prototyped and built a secure, scalable web platform for digitizing government documents during a 48-hour hackathon.
- Collaborated in a fast-paced team environment to deliver a functional solution under strict time constraints.

Education

2024 – present Kathmandu, Nepal	BSc (Hons) Computer Science with Artificial Intelligence <i>Sunway College (Awarded 50% Merit Scholarship)</i>
2021 – 2023 Lalitpur, Nepal	+2 in Science <i>Little Angels' College</i>

References

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